TSSM
Creating VCE Success

2025 Trial Examination

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PROCESSING LABEL HERE

Write your **student number** in the boxes above. **Letter**

Specialist Mathematics
Written examination 2

Units 3 & 4 – Question and Answer Book

VCE Examination

Reading time: **15 minutes**
Writing time: **2 hours**

Materials supplied

- Task book of 22 pages

Instructions

- Write your responses in English.
- You may keep this Task Book.
- Students are permitted to bring into the examination room: pens, pencils, highlighters, erasers, sharpeners and rulers
- Students are NOT permitted to bring into the examination room: blank sheets of paper and/or white out liquid/tape.
- An approved CAS calculator, one scientific calculator (if desired) and one bound reference are permitted in this examination.
- Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room.

Contents pages

Section A (20 questions, 20 marks) _____ 2-8
Section B (5 questions, 60 marks) _____ 8-22

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SECTION A – Multiple-choice questions
Instructions

- Answer **all** questions in pencil on the answer sheet provided for multiple-choice questions.
- Choose the response that is **correct** for the question.
- A correct answer scores 1, an incorrect answer scores zero.
- Marks will **not** be deducted for incorrect answers.
- No marks will be given if more than one answer is completed for any question.

Question 1

The equations of the asymptotes on the graph of the function: $y = \frac{1}{x-a} + \frac{2}{x^2+a}, a \in R^+$ are:

A. $x = a, x = -\sqrt{a}$ and $y = 0$
B. $x = a, x = -\sqrt{a}$ and $y = 3$
C. $x = a$ and $y = 0$
D. $x = a$ and $y = 3$

Question 2

Consider the function: $f: A \to R$ where $f(x) = \frac{2x}{\sin 2x - \cos 2x}$ where A is a suitable maximal domain. How many vertical asymptotes does the graph of function f have over $x \in [0, 2\pi]$?

A. 4
B. 5
C. 6
D. 7

Question 3

The contrapositive to the statement: **ALL CROWS ARE BLACK** is:

A. If it's not black, it's not a crow
B. No crows are black.
C. If it's black, it's a crow
D. All crows are not black.

Question 4

Consider the algorithm, written in pseudocode:

$for\ a\ from\ 1\ to\ 8$
$for\ b\ from\ 1\ to\ 12$
$for\ c\ from\ 1\ to\ 20$
$if\ a^2 + b^2 = c^2\ then$
$print\ (a, b, c)$
$end\ if$
$end\ for$
$end\ for$
$end\ for$

How many solutions will be in the printout?

A. 3
B. 4
C. 5
D. 6

Question 5

Which function, specified below might have a domain of: $\left[0, \frac{\pi}{2}\right] \setminus \frac{\pi}{4}$ and a range of $R \setminus (-2,4)$

A. $\mathrm{f}(x) = 3\sec 2x + 1$
B. $\mathrm{g}(x) = 3\mathrm{cosec}\, 2x + 1$
C. $\mathrm{h}(x) = 2\sec 3x + 1$
D. $\mathrm{m}(x) = 3\mathrm{cosec}\, 2x - 1$

Question 6

The cartesian equation of the curve defined by the parametric equations:

$x = 2\tan\left(\frac{t}{4}\right), y = \sec\left(\frac{t}{4}\right), -2\pi < t < 2\pi$ has asymptotes:

A. $y = \pm\frac{x}{2}$
B. $y = \pm 2x$
C. $y = \pm\frac{x}{4}$
D. $y = \pm 4x$

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Question 7

The gradient of the graph of $y = 2e^{2x+y} - 2$ when $y = 2$ is:

A. $\frac{4}{3}$

B. $-\frac{4}{3}$

C. $\frac{8}{3}$

D. $-\frac{8}{3}$

Question 8

Given $\operatorname{cosec} x = -1\frac{2}{3},\ x \in \left(\frac{3\pi}{2}, 2\pi\right)$, which statement is INCORRECT?

A. $\sec x = \frac{5}{4}$

B. $\tan x = -\frac{3}{4}$

C. $\sec 3x = -\frac{125}{44}$

D. $\sin 2x = \frac{24}{25}$

Question 9

Using Euler's approximation, given the initial condition: $x_0 = 0$ and $y_0 = 2$, find the value of y_2 for the differential equation $\frac{dy}{dx} = \log_e(\cos x)$, using step sizes of $\frac{\pi}{3}$.

A. $2 - \frac{\pi}{6}\log_e\left(\frac{\sqrt{3}}{2}\right)$

B. $y_2 = 2 + \frac{\pi}{3}\log_e(2)$

C. $y_2 = 2 - \frac{\pi}{3}\log_e(2)$

D. $y_2 = 2 - \frac{2\pi}{3}\log_e(2)$

Question 10

A circular puddle with an initial radius of 40 cm evaporates such that its area decreases by $20\ \text{cm}^2\ \text{min}^{-1}$. When, to the nearest hour, is the radius decreasing by $\frac{1}{\pi}\ \text{cm min}^{-1}$?

A. 3 hours

B. 4 hours

C. 5 hours

D. 6 hours

Question 11

Given $z = rcis(\theta)$ then, it is TRUE that:

A. $2z = 2r\text{cis}(2\theta)$

B. $z = -\bar{z}$

C. $\frac{1}{z} = r\text{cis}\left(\frac{1}{\theta}\right)$

D. $\frac{1}{z} = \frac{1}{r}\text{cis}(-\theta)$

Question 12

$z = 4 - i$ is **not** a solution over C to the equation:

A. $z^2 - 8z + 17 = 0$

B. $z^2 - 4z + 4i + 1 = 0$

C. $z^2 + 4z - 2iz - 4i - 1 = 0$

D. $z - 4 + i = 0$

Question 13

A circle with its centre at the point $z = 2 + 2i$ and passing through $z = -1 - 2i$ can be described by the relation:

A. $\{z: |z - 2 - 2i| = 5\}$

B. $\{z: |z - 2 - 2i| = 25\}$

C. $\{z: |z + 2 + 2i| = 5\}$

D. $\{z: |z + 1 + 2i| = 5\}$

Question 14

A surface of revolution is formed when $= \frac{6}{3-x}$, $1 \leq x \leq 2$, is revolved around the $x - axis$. The surface area in square units is closest to:

A. 101

B. 89

C. 32

D. 14

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Question 15

The acute angle between the straight lines specified by the vector equations:

$\underset{\sim}{r}_1(\lambda) = \underset{\sim}{i} + 2\underset{\sim}{j} - \underset{\sim}{k} + \lambda\left(2\underset{\sim}{i} - 2\underset{\sim}{j} + \underset{\sim}{k}\right), \lambda \in R$

$\underset{\sim}{r}_2(\mu) = 2\underset{\sim}{i} - \underset{\sim}{j} + 2\underset{\sim}{k} + \mu\left(\underset{\sim}{i} + \underset{\sim}{j} + 3\underset{\sim}{k}\right), \mu \in R$

is closest to:

A. 66^0
B. 68^0
C. 70^0
D. 72^0

Question 16

A particle with an initial position of 4 cm is moving in a straight line with a velocity of $2\sqrt{x}\ \text{cms}^{-1}$ Let x be the particle's position in cm, t seconds after beginning its motion. The acceleration ($a\ \text{cms}^{-2}$) and position of the particle after 2 seconds respectively are:

A. $a = 2\ \text{cms}^{-2}, x = 16\ \text{cm}$
B. $a = 2\ \text{cms}^{-2}, x = 4\ \text{cm}$
C. $a = 0.5\ \text{cms}^{-2}, x = 16\ \text{cm}$
D. $a = 0.5\ \text{cms}^{-2}, x = 4\ \text{cm}$

Question 17

Three independent, normally distributed random variables A, B and C are such that:

$E(A) = 4.8, Var(A) = 0.6, E(B) = 5.4, Var(B) = 0.3, E(C) = 7.1, Var(C) = 0.8.$

Given $D = A + 2B - C$, find correct to 3 decimal places, the probability that a random observation of D will be greater than 10 is:

A. 0.166
B. 0.176
C. 0.186
D. 0.196

Question 18

The life of SUPA batteries can be assumed to be normally distributed with a mean of μ months and a standard deviation of σ months. Based on a sample of 40 batteries, the 95% confidence interval, in months, for the life of a randomly selected battery is found to be $[72, 86]$

The 99% confidence interval for the life of a randomly selected battery is closest to:

A. $[69.2, 88.8]$
B. $[69.4, 88.6]$
C. $[69.6, 88.4]$
D. $[69.8, 88.2]$

Use the following information to help answer questions 19 *and* 20.

LUX light globes are known to have an average life of 640 hours with a standard deviation of 110 hours. A new light filament is trialled in order to test the hypothesis that the average life expectancy (of the 'new' globes) has increased significantly.

Question 19

A Type I and Type II error respectively would be:

A. The 'new' globes have a significantly higher average life when in fact they do not.
The 'new' globes do not have a significantly higher average life when in fact they do.

B. The original globes have a significantly higher average life when in fact they do not.
The original globes do not have a significantly higher average life when in fact they do.

C. The 'new' globes have a significantly higher average life and in fact they do.
The 'new' globes do not have a significantly higher average life and in fact they do not.

D. The 'new' globes have a significantly higher average life when in fact they do not.
The 'new' globes have a significantly higher average life when in fact they do.

Question 20

A sample of n 'new' globes are tested and found to have an average life of 672 hours. Assume the standard deviation remains at 110 hours. A one tailed hypothesis test at the 1% level of significance correctly concludes that there is no significant difference in average length of life between the original globes and the 'new' globes. The maximum number of globes tested is:

A. 59
B. 63
C. 67
D. 71

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SECTION B – Extended response questions
Instructions

- Answer **all** questions in the spaces provided.
- In **all** questions where a numerical answer is required, an exact value must be given unless otherwise specified.
- In questions where more than one mark is available, appropriate working **must** be shown.
- Unless otherwise indicated, the diagrams in this book are **not** to scale.

Question 1 (12 marks)
Particle *A* moves with a velocity of: $\underset{\sim}{v}_A(t) = \underset{\sim}{i} + (2t-6)\underset{\sim}{j}\ \text{ms}^{-1}, t \geq 0$

$\underset{\sim}{i}$ represents 1 metre in the $x-$ direction and $\underset{\sim}{j}$ represents 1 metre in the $y-$ direction.

a. Find the initial speed of particle *A*.

2 marks

b. Find when particle *A* is travelling parallel to the $x-$ direction and its velocity then.

2 marks

c. Find both the position vector and cartesian equation of particle *A* given it is initially at $\left(4\underset{\sim}{i} + 8\underset{\sim}{j}\right)$ m.

2 marks

The position of Particle B is: $\underset{\sim}{x_B}(t) = (at - 4)\underset{\sim}{i} + (a^2t - 8)\underset{\sim}{j}\ \ m, a \in R,\ t \geq 0$

$\underset{\sim}{i}$ represents 1 metre in the $x-$ direction and $\underset{\sim}{j}$ represents 1 metre in the $y-$ direction.

d. Find the value of a given the two particles collide.

2 marks

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e. Find when and where the two particles collide.

_____

_____

_____

_____

_____

_____

2 marks

f. Find to the nearest degree, the acute angle between the two particles at the point of collision.

_____

_____

_____

_____

_____

_____

2 marks

2 + 2 + 2 + 2 + 2 + 2 = 12 marks

Question 2 (11 marks)

BLUSH tomatoes grown in a nursery have heights that are normally distributed with a mean of $1.24\ \mathrm{m}$ and a standard deviation of $0.21\ \mathrm{m}$. A sample of 30 plants are given extra fertilizer (named BLUSH PLUS) are then monitored over an extended period. The resulting sample mean life $(\bar{X})$ and standard deviation, assumed to be normally distributed, are $1.31\ \mathrm{m}$ and $0.21\ \mathrm{m}$ respectively.

a. Find, correct to two decimal places, the 95% confidence interval for the actual mean life the BLUSH PLUS tomatoes.

2 marks

b. Find $E(\bar{X})$ and $sd(\bar{X})$ hence find, correct to 3 decimal places, the chance that the mean life of the sample of BLUSH PLUS tomatoes is greater than $1.31\ \mathrm{m}$ given the actual mean life is $1.24\ \mathrm{m}$.

3 marks

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Based on the sample results described above, nursery staff are discussing how significant the extra height in the BLUSH PLUS tomatoes is.

c. Write down an appropriate one-sided null hypothesis H_0 and an alternative hypothesis H_1 with regard to the sample average height compared to the original average height.

_____

_____

1 mark

d. Is the null hypothesis accepted or rejected at the 5% level of significance? Is the null hypothesis accepted or rejected at the 1% level of significance? Explain your answers.

_____

_____

_____

_____

_____

_____

2 marks

e. A new sample of 20 plants are given the extra fertilizer, producing a mean height of $\bar{Y}$. (Assume the standard deviation remains at 0.21 m.)

Given $\Pr(\bar{Y} > a \mid \mu = 1.24) = 0.2$, find $'a'$ correct to two decimal places.

3 marks

2 + 3 + 1 + 2 + 3 = 11 marks

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Question 3 (12 marks)

a. Given $z = x + yi$, expand z^2 writing your result in terms of $Re(z^2)$ and $Im(z^2)$.

2 marks

b. Given $(cis\ \theta)^2 = z^2$ write expressions for $\cos 2\theta$, $\sin 2\theta$ and $\tan 2\theta$ in terms of x and y.

3 marks

Suppose that $y = \sin\theta$ where $\sin 2\theta = \frac{4}{5}$

c. Show that the exact value of $\sin\theta$ can be found by solving the quartic equation:
$25y^4 + 15y^2 - 4 = 0$

3 marks

d. Find the exact value of $\sin\theta$.

1 mark

e. Use the expansion of $(\text{cis}\,\theta)^3 = z^3$ to find $\tan 3\theta$ as a simplified fraction.

3 marks

2 + 3 + 3 + 1 + 3 = 12 marks

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Question 4 (14 marks)

Consider the function $f: D \to R$ where $f(x) = \frac{2-2x}{\sqrt{4-x^2}}$

a. State the implied domain (D) of f.

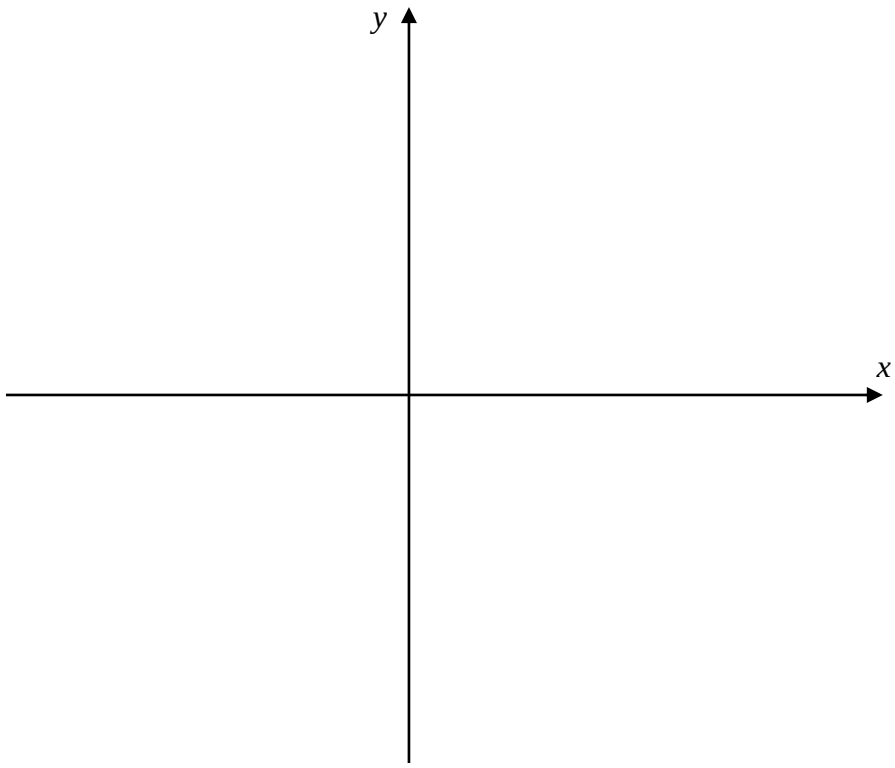
_____

_____

1 mark

b. Sketch $f: D \to R$ where $f(x) = \frac{2-2x}{\sqrt{4-x^2}}$ Label key features.

Shade the area bound by $y = f(x)$, the $x -$ axis and the $y -$ axis. (Area A).



3 marks

c. Using a suitable substitution and other calculus techniques, evaluate area A.
(Show all working.)

3 marks

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A steel cable used to stabilize a bridge is anchored at $(0,1)$ and $(1,0)$. Ideally the cable is fully tensioned and thus follows the straight line between these two anchor points. (Call this line $y = h(x)$.) In reality, the cable follows the line defined by $y = g(x)$ where:

$g: [0,1] \to R \quad g(x) = \frac{2-2x}{\sqrt{4-x^2}}$.

d. Find, correct to five decimal places, the length of the curve defined by:

$g: [0,1] \to R$ where $g(x) = \frac{2-2x}{\sqrt{4-x^2}}$.

2 marks

e. Expressed as a percentage, correct to three decimal places, how much longer the curve defined by $g: [0,1] \to R$ where $g(x) = \frac{2-2x}{\sqrt{4-x^2}}$ is than the straight-line length?

2 marks

f. Let $y = h(x)$ be the defined as the straight line starting at $(0,1)$ and ending at $(1,0)$. Find the volume of the solid formed V_x, when the region bound by $y = g(x)$ and $y = h(x)$ is rotated around the $x -$ axis. Give your answer correct to two decimal places.

3 marks

1 + 3 + 3 + 2 + 2 + 3 = 14 marks

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Question 5 (11 marks)

A stone is dropped from an air balloon. The balloon is 50 m above the ground and moving vertically up at a constant speed of 2 ms^{-1}. In parts $\boldsymbol{a}$. and $\boldsymbol{b}$. assume air resistance to be negligible.

a. Find when and with what speed the stone hits the ground. Give answers correct to 2 decimal places.

3 marks

A second stone, directly below the first stone is launched directly upwards from a height of 1 m, 1 s after the first stone is dropped. The second stone has an initial speed of 20 ms^{-1}. The two stones collide after T seconds.

b. Define two functions, x_1 and x_2, the positions of stones 1 and 2, t seconds after stone 1 is dropped. Where do the two stones collide? (Give your answer correct to two decimal places.)

3 marks

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c. Given instead the first stone experiences a deceleration due to air resistance of $0.1v^2\ \mathrm{ms}^{-2}$ find the maximum height of the stone and the speed it lands at. (Give your answers correct to two decimal places.)

5 marks

3 + 3 + 5 = 11 marks

End of examination questions